Supplementary Materials for: Observational evidence for more than a meter of 21st century sea-level rise

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Contents

1 Materials and Methods

Unless otherwise noted, all reported intervals are 90% Bayesian credible intervals (equal-tailed), and where 1σ is reported (e.g., the rheology correction factor), we state explicitly that it is a standard deviation.

1.1 Global mean sea level (GMSL) rate estimates

The satellite-era GMSL rates and 90% confidence intervals reported in Figure 1 of the main text are derived from the quadratic trajectory methodology of Hamlington et al. [1]. A quadratic is fit to the NASA satellite altimetry record (1993–2023) and the instantaneous rate at any time t is given by $\dot{H}(t) = a + bt$, where a and b are the linear and quadratic coefficients. The 90% confidence intervals are determined via a 10,000-member Monte Carlo simulation in which each realization perturbs the rate and acceleration by three independent error sources: (i) serially correlated residuals after removing the quadratic trajectory, following Maul & Martin [2]; (ii) glacial isostatic adjustment (GIA) uncertainty from a Bayesian inversion of ice loading histories; and (iii) satellite mission-specific measurement errors including altimeter noise, orbit determination, wet troposphere corrections, and inter-mission biases. The resulting 90% CIs at the endpoints of the record are 2.1 ± 1.0 mm yr^{-1} at 1993 and 4.5 ± 1.0 mm yr^{-1} at 2024.

The pre-satellite rate of 1.1 mm yr^{-1} (0.8–1.3 mm yr^{-1}, 90% CI) averaged over 1900–1940 is computed from the Frederikse et al. [3] tide-gauge reconstruction using a local polynomial regression (tricube kernel, 30-year bandwidth) that estimates instantaneous rates and their standard errors at each year. The narrower confidence interval reflects the longer averaging period and the lower sensitivity of tide-gauge uncertainties to the error sources that dominate the satellite record. Temperature changes between rate-doubling epochs are from Berkeley Earth [4].

1.2 Observational datasets

All component models are calibrated against published observational products (Table S1). We use:

- **Global mean sea level:** NASA satellite altimetry (TOPEX/Jason, 1993–present; Hamlington et al. [5]), with the Frederikse et al. [3] tide-gauge reconstruction (1900–2018) and Dangendorf et al. [6] reconstruction for pre-satellite validation.

- **Ocean thermal expansion:** NOAA thermosteric sea level (doi:10.7289/V53F4MVP), 0–700 m (1955–2025) and 0–2000 m (2005–2025) [7]. A 14% upward bias correction is applied to account for the documented low bias from vertical interpolation of sparse profiles [8]. Depth corrections for 700–2000 m (0.15 ± 0.05 mm yr^{-1}; [7, 9]) and below 2000 m (0.07 ± 0.04 mm yr^{-1}; [10, 11]) are applied cumulatively for years before full-depth data are available.

- **Glaciers:** GlaMBIE global consensus (2000–2023; GlaMBIE [12]), combining glaciological field measurements, satellite altimetry, DEM differencing, and GRACE/GRACE-

FO gravimetry across all 19 RGI regions. We use 2000–2023 (23 annual points) for calibration.

- **Greenland:** Mouginot et al. [13] SMB and discharge (1972–2018), with Mankoff et al. [14] (1986–2023) withheld for validation. Subsurface ocean temperature from EN4 v4.2.2 with g10 bias corrections (200–500 m, Greenland peripheral waters; Good et al. [15]). Surface temperature from Berkeley Earth [4].
- **Antarctica:** IMBIE satellite-era mass balance (1992–2020; Otosaka et al. [16]), partitioned by region.
- **Terrestrial water storage:** Frederikse et al. [3] reconstruction (1900–2018).
- **Temperature scenarios:** IPCC AR6 assessed GMST projections (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5; [17]).

Table S1: Observational and projection datasets used in the component models. A complete dataset inventory including versions, spatial domains, and processing details is provided in `data/dataset_inventory.xlsx`.

Component	Dataset	Time range	Reference
GMSL	NASA satellite altimetry	1993–present	Hamlington et al. [5]
GMSL	Frederikse GMSL reconstruction	1900–2018	Frederikse et al. [3]
Thermosteric	NOAA thermosteric SL (0–700 m)	1955–present	Levitus et al. [7]
Thermosteric	NOAA thermosteric SL (0–2000 m)	2005–present	Levitus et al. [7]
Glaciers	GlaMBIE global consensus	2000–2023	Zemp et al. [12]
Greenland	Mouginot SMB & discharge	1972–2018	Mouginot et al. [13]
Greenland (val.)	Mankoff PROMICE	1986–present	Mankoff et al. [14]
Greenland	EN4 subsurface ocean T (v4.2.2)	1970–2021	Good et al. [15]
Temperature	Berkeley Earth GMST	1850–present	Rohde & Hausfather [4]
WAIS	IMBIE West Antarctica	1992–2020	Otosaka et al. [16]
EAIS	IMBIE East Antarctica	1992–2020	Otosaka et al. [16]
Peninsula	IMBIE Antarctic Peninsula	1992–2020	Otosaka et al. [16]
TWS	Frederikse reconstruction	1900–2018	Frederikse et al. [3]
Projections	IPCC AR6 FACTS (med. & low conf.)	2020–2150	Garner et al. [18]

1.3 Component models

1.3.1 Thermosteric

We fit a two-layer energy balance model [19] jointly to NOAA in situ thermosteric sea level at 0–700 m (1955–2025, 71 annual points) and 0–2000 m (2005–2025, 21 points) [7]. The governing equations are:

$$\frac{dS_u}{dt} = \frac{T(t) - S_u}{\tau_u} \tag{1}$$

$$\frac{dS_d}{dt} = \frac{S_u - S_d}{\tau_d} \tag{2}$$

$$H_{\text{upper}}(t) = a\,S_u^2 + b_u\,S_u + c\,(t - t_0), \tag{3}$$

$$H_{\text{total}}(t) = H_{\text{upper}}(t) + b_d\,S_d, \tag{4}$$

where S_u and S_d are upper- and deep-ocean thermal state variables driven by GMST $T(t)$, τ_u is the upper-ocean relaxation time, and $\tau_d = 150\,\text{yr}$ is the deep-ocean relaxation time fixed at the CMIP5 multi-model mean [19]. A below-2000 m contribution of $0.07 \pm 0.04\,\text{mm}\,\text{yr}^{-1}$ [10, 11] is added as a post-hoc linear trend during projection (not included in calibration, as both depth ranges are demeaned before fitting). We apply a 14% upward correction to NOAA observations to account for the documented low bias from interpolation of sparse hydrographic profiles, primarily in the Southern Ocean [8], and carry a 5% residual structural uncertainty.

The six free parameters (a, b_u, b_d, c, $\log\tau_u$, $\log\sigma_{\text{extra}}$) are sampled via MCMC (emcee; 48 walkers, 4000 burn-in + 12000 production steps). The posterior median and 90% credible intervals are: $a = 19.3$ [9.5, 29.4] mm °C^{-2}, $b_u = 50.2$ [40.2, 72.3] mm °C^{-1}, $b_d = 67.8$ [36.8, 101.6] mm °C^{-1}, $c = 0.020$ [-0.110, 0.135] mm yr^{-1}, $\tau_u = 11.1$ [5.9, 23.7] yr, and $\sigma_{\text{extra}} = 2.0$ [1.7, 2.3] mm. The model achieves $R^2 = 0.965$ (0–700 m) and $R^2 = 0.978$ (0–2000 m).

Parameter identifiability. Because a and b_u multiply functions of S_u, which itself depends on τ_u, a potential concern is parameter degeneracy: changing τ_u rescales S_u, which could in principle be compensated by rescaling a and b_u. Under full degeneracy, $b_u \propto \tau_u$ and $a \propto \tau_u^2$. The posterior does not exhibit this: τ_u varies by a factor of 3.9 across the 90% CI, while b_u varies by only a factor of 1.8 (not 3.9) and a by a factor of 3.1 (not 15). This partial independence arises from four sources: (1) the joint fit to two depth ranges with different time spans, where the deep-ocean response $b_d\,S_d$ provides an independent constraint on τ_u through the $S_u \to S_d$ coupling; (2) the quadratic term $a\,S_u^2$, whose shape (curvature) changes with τ_u and cannot be compensated by a simple rescaling of a; (3) the interannual and decadal variability in GMST, which constrains the phase lag of S_u relative to $T(t)$ independently of amplitude; and (4) the trend term $c\,(t - t_0)$, which is entirely τ_u-independent. Pairwise posterior correlations are moderate ($|r| < 0.7$ for all parameter pairs), and no bimodal structure is evident in the posterior. Corner plots are provided in the analysis notebook.

The ODE system is solved via an exponential integrator (piecewise-constant midpoint forcing, analytical decay) with initial conditions $S_u(0) = S_d(0) = T(0)$ (equilibrium assumption). To generate projections, we drive the fitted model with IPCC AR6 SSP temperature trajectories, propagating posterior parameter uncertainty, SSP temperature trajectory uncertainty, and below-2000 m rate uncertainty ($0.07 \pm 0.04\,\text{mm}\,\text{yr}^{-1}$) via 2,000 Monte Carlo samples.

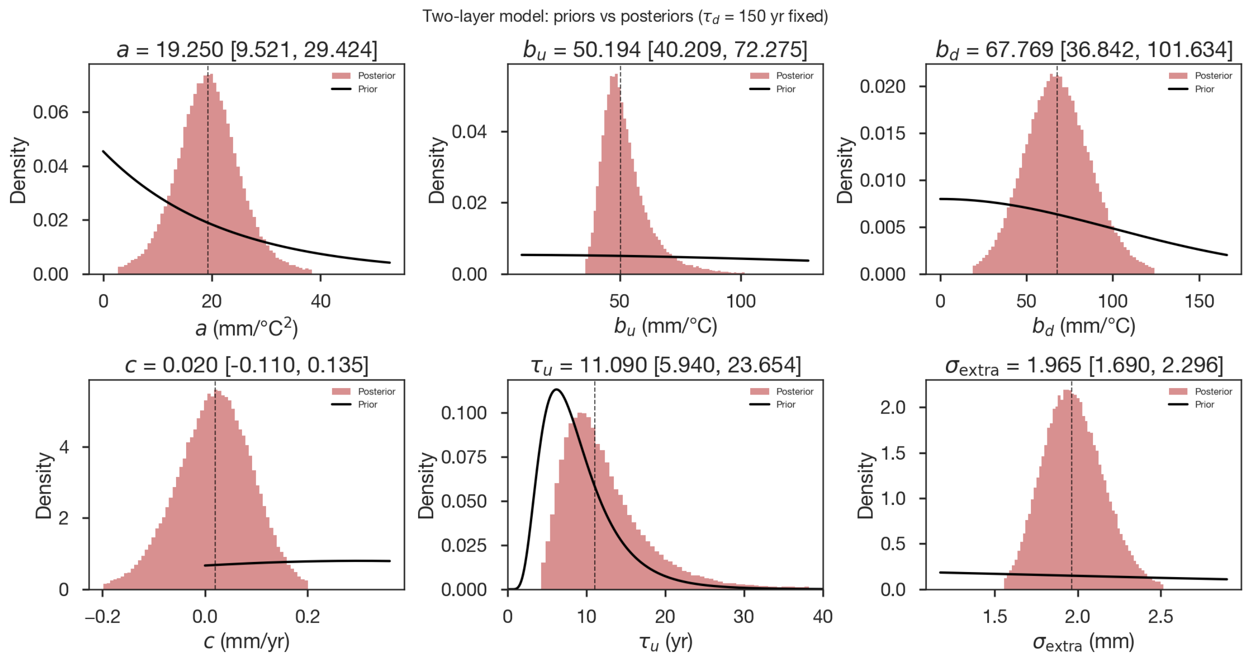

Figure S1: **Two-layer thermosteric model: priors and posteriors.** Prior distributions (black curves) and posterior histograms (teal) for the six free parameters of the two-layer energy balance model fitted jointly to NOAA 0–700 m and 0–2000 m thermosteric sea level. Dashed vertical lines indicate the posterior median; panel titles report the median and 90% credible interval. The deep-ocean relaxation time $\tau_d = 150\,\text{yr}$ is fixed. For all parameters the posteriors are substantially narrower than and displaced from the priors, indicating that the observations—not the prior assumptions—are driving the inference. The one exception is b_d, whose posterior is wider than the others, reflecting the shorter 0–2000 m record (21 years vs 71 years for 0–700 m).

1.3.2 Glaciers

We model cumulative glacier sea-level contribution in level space as

$$H_{\text{gl}}(t) = a\int_{t_0}^{t} T^2\,d\tau \;+\; b\int_{t_0}^{t} T\,d\tau \;+\; c\,(t-t_0) \;+\; H_0 \;+\; \varepsilon(t), \tag{5}$$

where T is the GMST anomaly, the integrals are computed on a monthly grid and evaluated at annual observation times, H_0 is a baseline offset, and $\varepsilon(t) \sim \mathcal{N}(0,\,\sigma^2_{\text{obs}} + \sigma^2_{\text{extra}})$ with σ_{extra} an additional model-inadequacy noise term learned from the data. The corresponding rate model is $\dot{H}_{\text{gl}} = a\,T^2 + b\,T + c$. We fit both the quadratic (5 free parameters: a, b, c, σ_{extra}, H_0) and linear (4 free parameters, with $a = 0$) models to the GlaMBIE record (2000–2023, 23 annual points) using Bayesian inference with weakly informative priors (emcee; linear fit: 32 walkers, 2000 burn-in + 4000 production steps, thinning by 2; quadratic fit: 64 walkers, 4000 burn-in + 12000 production steps, thinning by 2). BIC model selection favors the linear model ($\Delta\text{BIC} = \text{BIC}_{\text{lin}} - \text{BIC}_{\text{quad}} = -63.8$), constituting decisive evidence for the more parsimonious linear model. The data detect nonzero curvature—the posterior probability $P(a > 0) > 99\%$ with $a = 0.009\,\text{mm}\,\text{yr}^{-1}\,{}^\circ\text{C}^{-2}$—but the quadratic coefficient is small ($a \ll b$)

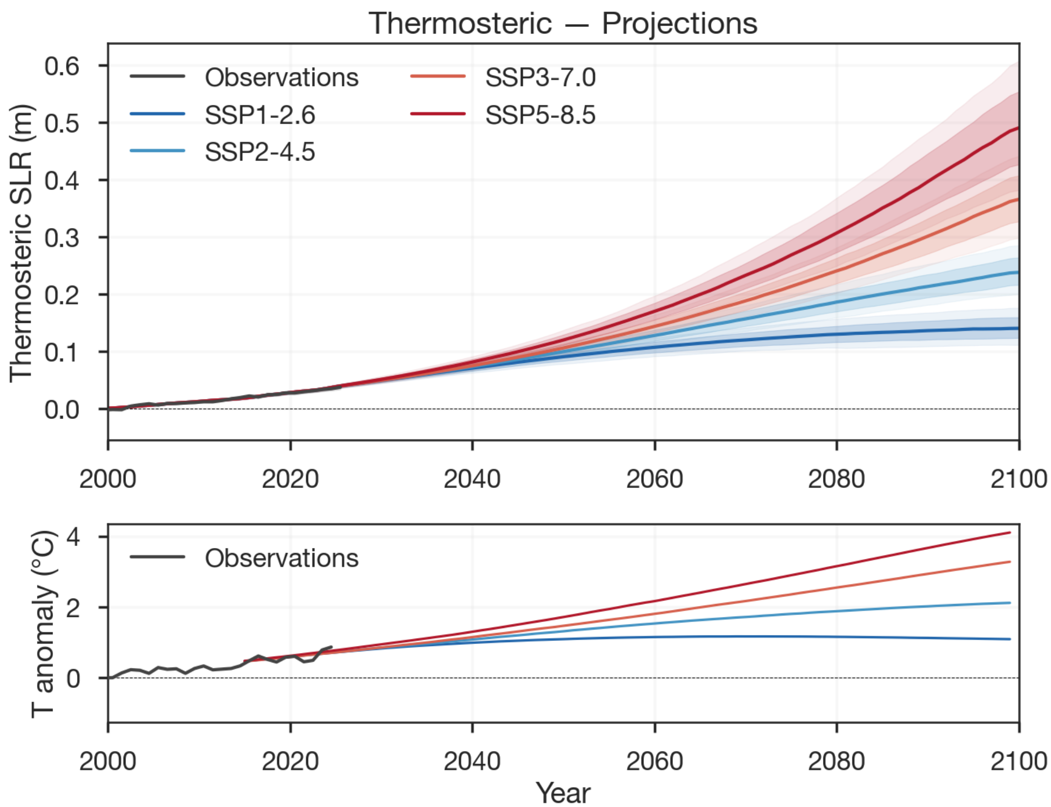

Figure S2: **Thermosteric sea-level projections.** Upper panel: median and 66%/90% credible intervals for the two-layer model under four SSP scenarios, with full-depth observations (black). Lower panel: GMST forcing trajectories (relative to year 2000) with Berkeley Earth observations.

and BIC favors the linear model, so we adopt the linear model for projections. The posterior median sensitivity is $b = 0.66$ mm yr^{-1} °C^{-1} and the committed rate is $c = 0.57$ mm yr^{-1}, with $R^2 = 0.995$. Cumulative loss is capped at the total glacier volume of 0.32 m SLE [20].

1.3.3 Greenland SMB

We use RACMO-derived SMB [13] for the hindcast. Forecasts use a quadratic temperature scaling $\Delta\dot{H}_{\rm smb} = C_T\,\Delta T + C_{T^2}\,\Delta T^2$ with sensitivities drawn from the inter-RCM spread [21–23]. The linear sensitivity $C_T \sim \mathcal{N}(-300, 80^2)$ Gt yr^{-1} °C^{-1} of GMST encompasses the MAR, RACMO, and HIRHAM estimates. The quadratic sensitivity is $C_{T^2} \sim \mathcal{N}(-50, 30^2)$ Gt yr^{-1} °C^{-2}. The cumulative SMB contribution is obtained by integrating the rate anomaly:

$$\Delta H_{\rm smb}(t) = \int_{t_0}^{t} \left[\dot{M}_0 + C_T\,\Delta T(t') + C_{T^2}\,\Delta T(t')^2\right] dt', \tag{6}$$

where ΔT is the GMST anomaly relative to the 1995–2005 baseline and $\dot{M}_0 = 380$ Gt yr^{-1} is the baseline SMB.

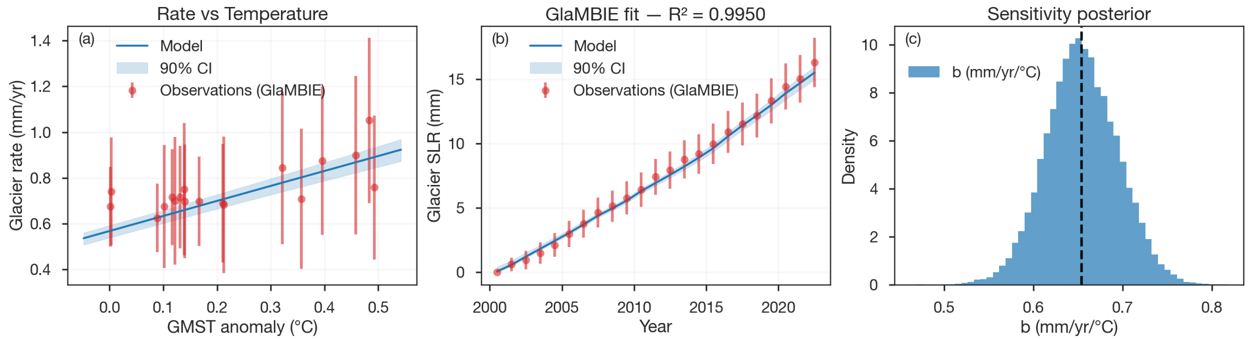

Figure S3: **Glacier model diagnostics.** (a) Glacier mass loss rate vs GMST anomaly with the fitted linear relationship. (b) Cumulative glacier SLR: GlaMBIE observations (circles) with the linear model fit (black line) and 90% credible interval (grey shading); $R^2 = 0.995$. (c) Posterior distribution for the temperature sensitivity b.

1.3.4 Greenland discharge

The cumulative discharge contribution is modeled as the time-delayed integral of subsurface ocean warming:

$$\Delta H_{\mathrm{dyn}}(t) = \gamma \int_{t_0}^{t} T_{\mathrm{ocean}}(t' - \delta)\, dt' + r_0\,(t - \bar{t}), \tag{7}$$

where γ is the discharge sensitivity to ocean temperature, δ is the time delay between ocean warming and the ice sheet response, and r_0 is the background discharge rate at the baseline ocean temperature. T_{ocean} is from EN4 v4.2.2 with g10 bias corrections [15], averaged over 200–500 m depth in Greenland peripheral waters (58–80°N, 75–5°W). For each candidate delay $\delta \in \{4, 5, 6, 7, 8\}$ yr, we fit (γ, r_0) by weighted least squares against demeaned cumulative discharge from Mouginot et al. [13], with an additional rate constraint at the end of the record. Posterior samples are drawn from a BIC-weighted Gaussian mixture over the per-δ WLS covariance matrices, yielding $\gamma = 0.365$ mm SLE yr^{-1} °C^{-1} (90% CI: 0.354–0.377 mm SLE yr^{-1} °C^{-1}), $r_0 = 1.383$ mm SLE yr^{-1} (90% CI: 1.379–1.387 mm SLE yr^{-1}), and $\delta = 5$ yr (rate-space $R^2 = 0.794$).

We derive a simple scaling to contextualize the inferred value $\gamma = 0.37$ mm SLE yr^{-1} °C^{-1}. Taking the time derivative of Eq. (7), the discharge rate anomaly is

$$\dot{D} = \gamma\, T_{\mathrm{ocean}}(t - \delta), \tag{8}$$

so γ is the change in discharge rate per degree of subsurface ocean warming. The physical chain linking ocean temperature to discharge is: (i) ocean thermal forcing drives submarine melt at tidewater glacier termini, (ii) melt-driven undercutting promotes calving and terminus retreat, and (iii) the resulting reduction in resistive stress at the terminus increases ice flux. These steps can be combined into a scaling for the aggregate discharge sensitivity γ.

Submarine melt. For a single outlet glacier, the submarine melt rate scales approximately linearly with ocean thermal forcing above a reference temperature T_f:

$$\dot{m} = \kappa\,(T_{\mathrm{ocean}} - T_f), \tag{9}$$

where $\kappa \sim 10$–100 m yr^{-1} °C^{-1} depending on fjord geometry, subglacial discharge, and circulation [24–26].

Dynamic amplification. Submarine melt thins and undercuts the terminus, triggering calving and grounding-line retreat that increase ice flux by a factor μ beyond the direct melt loss. For Greenland's tidewater glaciers, observations and models suggest $\mu \sim 3$–10 [27, 28]. The total mass flux perturbation from a single glacier of effective calving-front cross-section A_f is then

$$\Delta\dot{Q}_i \sim \mu\,\rho_{ice}\,A_{f,i}\,\kappa_i\,\Delta T_{\text{ocean}}, \tag{10}$$

where ρ_{ice} is the mass density of ice.

Aggregation. Summing over Greenland's $N \approx 200$ marine-terminating outlets:

$$\gamma_{\text{phys}} = \frac{1}{A_{\text{ocean}}\,\rho_w}\sum_{i=1}^{N}\mu_i\,\rho_{ice}\,A_{f,i}\,\kappa_i, \tag{11}$$

where $A_{\text{ocean}} = 3.625\times10^{14}$ m^2 is the ocean area and ρ_w is seawater density. Using representative values ($\bar{\kappa} \sim 30$ m yr^{-1} °C^{-1}, $\bar{\mu} \sim 5$, mean cross-section $\bar{A}_f \sim 0.15$ km^2, $\rho_{ice} = 900$ kg m^{-3}) gives

$$\gamma_{\text{phys}} \sim 0.22\text{–}0.55\ \text{mm SLE yr}^{-1}\,{}^\circ\text{C}^{-1}, \tag{12}$$

which brackets the calibrated $\gamma = 0.37$ mm SLE yr^{-1} °C^{-1}.

As an independent check, over 1997–2018, Mouginot et al. [13] observed a discharge increase of $\Delta D{\sim}0.19$–0.28 mm SLE yr^{-1} while EN4 subsurface temperatures (200–500 m) in Greenland peripheral waters warmed by ∼0.5–1 °C [15], giving an observed bulk sensitivity of 0.19–0.55 mm SLE yr^{-1} °C^{-1}. The calibrated γ at the median ocean warming of $\Delta T{\sim}0.7$ °C predicts a rate increase of $\gamma = \Delta D/\Delta T \approx 0.37$ mm SLE yr^{-1} °C^{-1}, consistent with both the scaling and the observed acceleration.

Surface-to-ocean transfer function. To project T_{ocean} into the future, we calibrate a linear transfer function on annual-mean observations:

$$T_{\text{ocean}}(t) = \alpha\,T^{\text{Gr}}_{\text{surface}}(t) + \beta + \varepsilon(t), \tag{13}$$

where $T^{\text{Gr}}_{\text{surface}}$ is the Greenland regional surface temperature anomaly (Berkeley Earth gridded, annual mean), and ε is a zero-mean residual. Both series are anomalies relative to 1995–2005. An OLS fit over 52 common annual values gives $\alpha = 0.148 \pm 0.027$, $\beta = -0.051 \pm 0.031$ °C, $R^2 = 0.38$, and residual standard deviation $\sigma_\varepsilon = 0.20$ °C. The transfer function is calibrated independently of the discharge model parameters (γ, r_0, δ): the two regressions share no free parameters and are fit on different dependent variables (T_{ocean} vs. cumulative discharge).

For projections, SSP GMST trajectories are converted to Greenland regional surface temperature using an Arctic amplification factor (AA) of 3.0 ± 0.5 (1σ), consistent with the observed ratio of Greenland-to-global warming over 1970–2021. SSP Greenland regional surface temperatures (CMIP6, rebaselined to Berkeley Earth over 1995–2005) are passed through Eq. (13) to obtain projected T_{ocean}, which then enters Eq. (7) via the delayed integral. Each of the 2,000 Monte Carlo samples draws AA $\sim \mathcal{N}(3.0, 0.5^2)$, $\alpha \sim \mathcal{N}(0.148, 0.027^2)$, and $\beta \sim \mathcal{N}(-0.051, 0.031^2)$ jointly with (γ, r_0, δ) from the discharge posterior, so that Arctic amplification, transfer-function, and discharge parameter uncertainties all propagate into the projections. A single draw of $(\alpha, \beta, \gamma, r_0, \delta)$ is applied to all SSP scenarios, so that inter-SSP differences isolate the effect of the emissions pathway from parameter uncertainty.

1.3.5 EAIS

The EAIS component uses the same level-space Bayesian framework as glaciers and the Peninsula. The rate–temperature model is

$$\dot{H}_{\text{eais}}(t) = a\,T(t)^2 + b\,T(t) + c, \tag{14}$$

where $T(t)$ is the GMST anomaly. This is fit to the IMBIE EAIS record (1992–2020, 29 annual points; [16]) via emcee (32 walkers, 2000 burn-in + 4000 production steps, thinning by 2). The quadratic model yields $R^2 = 0.66$ (posterior mean: $a = 0.054$, $b = 0.035$, $c = -0.023$ mm yr^{-1}), and the linear model yields $R^2 = 0.56$ ($b = 0.021$ mm yr^{-1} °C^{-1}, $c = -0.018$ mm yr^{-1}). BIC weakly favors the quadratic (ΔBIC $= +4.1$ (BIC$_{\text{lin}}$ − BIC$_{\text{quad}}$, same convention as above)), but both R^2 values are too low for reliable projections given the 29-year record and large interannual variability (Fig. S7). EAIS projections instead use the literature-derived accumulation sensitivity $C_T = 60 \pm 20$ Gt yr^{-1} °C^{-1} of GMST [29, 30], reflecting the Clausius–Clapeyron scaling of snowfall with warming ($C_T > 0$ corresponds to mass gain, i.e., negative SLR). The EAIS contribution is small (−13 to −30 mm at 2100, depending on scenario).

1.3.6 Peninsula

The Antarctic Peninsula component uses the same level-space Bayesian framework, with the rate–temperature model $\dot{H}_{\text{pen}}(t) = a\,T(t)^2 + b\,T(t) + c$ calibrated against the IMBIE Peninsula record (1992–2020, 29 annual points; [16]) via emcee (32 walkers, 2000 burn-in + 4000 production steps, thinning by 2). BIC favors the linear model (ΔBIC $= -3.5$ (BIC$_{\text{lin}}$ − BIC$_{\text{quad}}$)). The adopted linear model achieves $R^2 = 0.94$ with posterior median sensitivity $b = 0.055$ mm yr^{-1} °C^{-1} and committed rate $c = 0.028$ mm yr^{-1} (Fig. S8). The Peninsula contribution reflects combined SMB and discharge responses to warming and is small relative to other components (∼7–12 mm at 2100).

1.3.7 Terrestrial water storage

TWS projections are adopted directly from the IPCC AR6 `landwaterstorage` component rather than independently modeled, as the processes governing groundwater depletion, reservoir impoundment, and land hydrology changes are outside the scope of this framework. At each decadal interval in the IPCC projections, we fit a Gaussian distribution to the reported median and inter-quantile spread and draw 2,000 Monte Carlo samples, which are then linearly interpolated to annual resolution. For the hindcast period (pre-2020), we use the Frederikse et al. [3] TWS observations with Monte Carlo sampling from their reported uncertainties. TWS is SSP-dependent but the contribution is small (∼30 mm at 2100 under SSP2-4.5) and does not materially affect the total uncertainty budget.

1.3.8 WAIS

Ice dynamics governing WAIS mass loss involve instability mechanisms that current process models have not proven themselves capable of fully representing [31, 32]. In the absence of reliable physical models, we take arguably the only approach available by constructing a scenario-mixture framework that represents this deep uncertainty, is grounded in the scholarly literature, and provides an accounting for recent discoveries, like the higher value of the viscous stress-exponent n [33, 34] and sensitivity of sea-level rise projections to that value [35, 36]. Our framework comprises three physically motivated scenarios, each characterized by a probability weight, an endpoint distribution at 2100, a temporal trajectory, and a multiplicative rheology (n) correction. We choose to set all WAIS scenarios to be independent of the warming trajectory because there are vanishingly few data to constrain the sensitivity of WAIS to ocean thermal forcing in a time when large calving events are rapidly changing ice shelves—the floating extension of the ice sheet—and because evidence suggests ocean thermal forcing sufficient to sustain WAIS retreat is committed regardless of emissions pathway [37]. We recognize that WAIS likely retains some sensitivity to emissions pathway, but that sensitivity is likely to be small relative to the wide uncertainties arising from dynamics already underway.

Observational context

The IMBIE satellite-era record (1992–2020; [16]) shows WAIS cumulative mass loss of $\sim$6.6 mm SLE, with the rate increasing from $\sim$0.15 $\mathrm{mm\,yr^{-1}}$ in 1992 to a peak of $\sim$0.44 $\mathrm{mm\,yr^{-1}}$ around 2010–2012. The mass loss signal emerges from measurement noise (signal-to-noise ratio > 2) by approximately 1998. A structural break at $\sim$2010 separates a period of acceleration (0.015 $\mathrm{mm\,yr^{-2}}$, 1992–2010) from a period of apparent deceleration (-0.031 $\mathrm{mm\,yr^{-2}}$, 2010–2020). This recent deceleration may reflect internal ocean variability in the Amundsen Sea [38] rather than a reversal of the ice-dynamical trend. The WAIS contribution represents $\sim$8–9% of the total GMSL rate and $\sim$35–70% of the barystatic (non-thermosteric) contribution.

Few if any runs in the 188-member ISMIP6 ensemble [31] reproduce the observed trajectory of WAIS mass loss through 2020 (Fig. 2f of the main text). The ensemble 95th percentile merely tracks the lower bound of observations, and the ensemble median implies near-zero or slightly positive mass change under warming—a result that is difficult to reconcile with satellite measurements and reflects known structural biases in the underlying models (see "Known biases in ice sheet model projections" below).

Framework structure: three scenarios

The framework comprises three scenarios spanning the range from stable WAIS (no instability) through marine ice sheet instability (MISI) to the combination of MISI with marine ice cliff instability (MICI):

- **S1: Status quo** ($P = 0.10$; 25–85 mm at 2100; $\alpha = 0$). No marine ice sheet instability operates. Discharge responds linearly to ocean thermal forcing, and grounding-line retreat does not trigger the positive feedback characteristic of MISI. The range is derived from a physical scaling: the lower bound extrapolates the IMBIE time-averaged rate (0.24 mm yr^{-1}) over 80 yr from the anchor year (∼19 mm); the 25 mm lower bound rounds up to account for measurement uncertainty in the baseline rate; the upper bound applies the projected Amundsen Sea ocean warming enhancement factor ($f = [(2.15 + \Delta T_{\text{warm}})/2.15]^2 \approx 2.9$ for +1.5 °C warming; [37]) to the peak observed rate, yielding a linear ramp to ∼85 mm.

- **S2: MISI** ($P = 0.80$; 150–1000 mm at 2100; $\alpha = +4$). Marine ice sheet instability drives accelerating grounding-line retreat on retrograde beds, with amplification from processes that current models omit (calving, subglacial hydrology, ice damage). The positive skewness ($\alpha = +4$) is motivated by the analytical and numerical results of Robel et al. [39], who showed that the grounding-line flux nonlinearity ($q_g \propto h_g^{n+1}$, where h_g is grounding-line thickness; [40]) amplifies and skews the uncertainty distribution toward greater ice loss when the nonlinearity exponent exceeds 3 (realistic values are 4–5). In their 500-member Thwaites ensemble, fractional projection uncertainty ranges from 0.25 to 0.50 depending on forcing persistence, with systematic positive skew. The trajectory follows a power-law ramp $H(t) = H_{\text{anchor}} + H_{\text{remaining}} \cdot [(t - t_a)/(2100 - t_a)]^{\beta}$, where the trajectory exponent β is drawn from a log-normal distribution with median 1.8 and log-standard deviation 0.3, derived from the Schoof [40] grounding-line flux scaling as described in the "Temporal trajectory" section below.

- **S3: MISI + MICI** ($P = 0.10$; 600–1300 mm at 2100; $\alpha = -3$). Full instability cascade including marine ice cliff instability, in which loss of buttressing ice shelves exposes ice cliffs that exceed the yield strength of ice, triggering structural failure and rapid retreat [41, 42]. The S3 bounds are anchored to the IPCC AR6 low-confidence Antarctic ice sheet projection under SSP5-8.5 [43]: the lower bound (600 mm) corresponds to the 83rd percentile (559 mm, rounded), and the upper bound (1300 mm) corresponds to the 95th percentile (1309 mm, rounded). This anchoring is appropriate because the AR6 low-confidence projection incorporates the deep-uncertainty processes (MISI + MICI) that define S3. For GMSL purposes in this study, it is immaterial whether ice loss originates from WAIS or from marine basins of EAIS—both contribute to the same global ocean—and in the AR6 low-confidence storyline, WAIS dominates but EAIS marine basins also contribute. The upper bound additionally matches the Amundsen Sea Embayment ice volume above flotation ($V_{af} \approx 1100$ mm; [44]), providing an independent cross-check estimate. The negative skewness ($\alpha = -3$) in log-space concentrates probability toward the lower portion of the range, reflecting the assessment that if MICI operates, it is unlikely to do so at maximum efficiency; the limited weight ($P = 0.10$) separately captures the evidence against efficient MICI operation during the 21st century (see below). The trajectory exponent has median 2.2, reflecting the more rapid acceleration expected under combined instabilities.

Scenario weights

The 10–80–10 weight split reflects the following assessments:

S1 = 10% (no MISI). Sustained WAIS stability requires grounding lines to stabilize despite ongoing retreat on retrograde slopes, committed ocean warming [37], and observed acceleration. IPCC AR6 states that MISI "may be underway" in the Amundsen Sea Embayment (medium confidence; [43]); Joughin et al. [45] concluded that early-stage collapse of Thwaites is already underway; and van den Akker et al. [46] showed that present-day ocean thermal forcing, held constant, is sufficient to deglaciate large parts of WAIS within 300–2500 yr. The 10% weight reflects the small but non-zero possibility that observed trends are reversible or dominated by internal variability.

S2 = 80% (MISI). MISI is the most physically supported instability mechanism, included in IPCC medium-confidence projections [43], and is arguably already initiated [45, 47]. Ritz et al. [48] found MISI to be the dominant instability pathway in a probabilistic treatment of Antarctic contributions; Bamber et al. [49] structured expert judgment produced WAIS distributions dominated by MISI-type dynamics with strongly positive skew; and DeConto et al. [50] obtained total Antarctic median ∼340 mm under RCP8.5 without MICI but with nonlinear acceleration above 3 °C. The 80% weight reflects a consensus position that MISI is the baseline expectation for WAIS evolution.

S3 = 10% (MISI + MICI). IPCC AR6 assigns low confidence to MICI and excludes it from medium-confidence projections [43]. Several studies have challenged the efficiency of MICI during the 21st century: Morlighem et al. [51] found that dynamic thinning prevents cliffs from growing tall enough for sustained structural failure, requiring calving rates $\geq 25\times$ higher than physically motivated models suggest; Clerc et al. [52] showed that realistic ice-shelf removal timescales (days or longer) raise the critical cliff height from ∼90 m to ∼540 m via viscous relaxation; Schlemm et al. [53] demonstrated that mélange buttressing and embayment geometry changes are self-limiting; and a factor-of-2 error in the derivation of critical cliff height arising from an incorrect application of the deviatoric stress decomposition [cf. Eq. 1 of 41] calls into question the prescribed value of the critical cliff height used in projections [42]. However, if our goal is to represent the scholarly literature without regard for individual views and opinions, then MICI cannot be entirely excluded. Our 10% weight is consistent with AR6's low-confidence assessment. We note that our conclusions are qualitatively insensitive to MICI because of the low-confidence weighting of 10%.

Statistical model for endpoint distributions

Within each scenario, the 2100 endpoint is drawn from a skew-normal distribution in log-space, $\mathrm{SN}_{\log}(\xi, \omega, \alpha)$, where ξ and ω are solved such that the 5th and 95th percentiles match the scenario bounds. The log-space formulation is motivated by three physical considerations: (i) the dominant uncertainties (bed geometry, basal friction, ocean thermal forcing, buttressing loss) enter as multiplicative factors on grounding-line flux $q_g \propto h_g^{n+1}$ [40], making a multiplicative (log-space) distribution natural; (ii) mass loss is strictly positive, which log-space enforces without truncation; and (iii) the scenarios span two orders of magnitude (25–1300 mm), and a log-space family provides scale-invariant behavior across this range. For $\alpha = 0$, the distribution reduces to a log-normal.

The skew-normal extension captures the asymmetry arising from flux nonlinearity. Robel et al. [39] showed analytically that when the grounding-line flux exponent $n+1 > 3$, ensemble distributions of retreating grounding lines are systematically skewed toward greater retreat: more-retreated members retreat faster (positive feedback), while less-retreated members experience weaker instability, creating a heavy right tail.

Temporal trajectory

The cumulative WAIS contribution at year t is given by

$$H(t) = H_{\text{anchor}} + (H_{2100} - H_{\text{anchor}}) \left(\frac{t - t_a}{2100 - t_a} \right)^{\beta}, \tag{15}$$

where H_{anchor} is the IMBIE observed cumulative value at the anchor year $t_a = 2020$ (the final year of the IMBIE record; $H_{\text{anchor}} = 5.8 \pm 0.7$ mm, 1σ, relative to the year-2000 baseline), H_{2100} is the scenario endpoint, and β is a trajectory exponent drawn from a log-normal distribution. For years before the anchor ($t \leq t_a$), the framework returns IMBIE observations with their measurement uncertainties rather than scenario projections. For S1 (no instability), $\beta = 1$ (constant rate, linear ramp). For S2, β has log-mean $\ln(1.8)$ and log-standard deviation 0.3, reflecting the accelerating, back-loaded character of MISI dynamics. The power-law trajectory form is motivated by the Schoof [40] grounding-line flux scaling $q_g \propto h_g^{n+1}$: if grounding-line retreat proceeds at roughly constant rate on a retrograde bed with slope s, then h_g decreases approximately linearly, $h_g(t) \sim h_0 - sv(t - t_0)$, and the cumulative flux integral grows faster than linearly because q_g increases as h_g decreases. For a power-law flux, dimensional analysis gives cumulative loss scaling as $(t - t_0)^{\beta}$ with $\beta > 1$, where the exact value depends on bed geometry and n; for $n = 3$–4, this yields $\beta \approx 1.5$–2.5. For S3, β has log-mean $\ln(2.2)$ and log-standard deviation 0.3, reflecting the more rapid acceleration expected when MICI supplements MISI.

The trajectory exponent is corrected for rheology: the reference β is drawn assuming $n = 3$ (matching the literature from which the scaling is derived) and then adjusted to $\beta_{\text{corrected}} = \beta_{n=3} \cdot (n_{\text{draw}} + 1)/(n_{\text{ref}} + 1)$, where $n_{\text{draw}} \sim \mathcal{N}(4.1, 0.4^2)$ is drawn from the observed Glen's law exponent distribution [33, 34]. This correction increases β by $\sim$28% at the median, producing more back-loaded (accelerating) trajectories when the ice viscosity has a higher sensitivity to stress than commonly assumed with $n = 3$.

Rheology correction

A multiplicative rheology correction is applied to all scenario endpoints. All ISMIP6 models and the literature scenarios from which our endpoint ranges are derived assume Glen's flow law with $n = 3$, but satellite observations indicate $n = 4.1 \pm 0.4$ in fast-flowing Antarctic ice [33], supported by laboratory measurements ($n \approx 3.5$–3.7; [54]), Greenland observations ($n \approx 4$; [55]), and theoretical analysis of the dislocation creep regime [34]. The consequence for projections has been quantified: Martin et al. [36] found that using $n = 3$ when the true value is $n = 4$ underestimates 100-year sea-level rise by 21–35% in the MISMIP+ and ABUMIP benchmarks, and Getraer & Morlighem [35] found $32 \pm 14\%$ greater Amundsen Sea Embayment loss with $n = 4$ in an ISMIP6-framework experiment. The bias grows nonlinearly with forcing strength [36].

We implement the rheology correction in Mode B (unified n-driven corrections; defined as one option in the accompanying jupyter notebook), in which a single draw of the Glen's law exponent n per Monte Carlo sample drives both corrections through the physical link to n:

1. $n_{\text{draw}} \sim \mathcal{N}(4.1, 0.4^2)$.

2. Endpoint: $R(n) = 1 + r_0\,(n_{\text{draw}} - n_{\text{ref}})$, where $r_0 = 0.28 \pm 0.07$ (1σ) is the fractional change in endpoint per unit change in n, calibrated from Martin et al. [36] and Getraer & Morlighem [35].

3. Trajectory: $\beta(n) = \beta_{\text{ref}} \cdot (n_{\text{draw}} + 1)/(n_{\text{ref}} + 1)$.

This couples the endpoint and trajectory corrections through n, which is more physically consistent than drawing R and the β correction independently (Mode A, defined in the accompanying jupyter notebook). At the median observed exponent $n = 4.1$, $R = 1 + 0.28 \times 1.1 = 1.31$; at $\pm 1\sigma$ in n, the correction spans from $R = 1.20$ ($n = 3.7$) to $R = 1.42$ ($n = 4.5$), consistent with the literature range of 1.21–1.35 from Martin et al. [36]. The correction is applied to all scenarios because the rheology bias affects every ice sheet model regardless of which instability mechanism operates. Importantly, R and β correct conceptually independent quantities: the endpoint ranges are defined at $n = 3$, and R scales the total ice lost (the endpoint magnitude), while the β correction changes when that ice is lost (the trajectory shape). These corrections are not double-counting because Martin et al. [36] and Getraer & Morlighem [35] report the ratio of total sea-level change at $n = 4$ to $n = 3$—a cumulative endpoint quantity—not the trajectory shape.

Known biases in ice sheet model projections

Our scenario ranges exceed IPCC AR6 medium-confidence projections for WAIS. This is expected and physically justified by multiple documented biases in the ISMIP6 ensemble:

1. *Rheology bias.* All ISMIP6 models assume $n = 3$; observed $n \approx 4$ produces 21–35% more sea-level rise [35, 36]. Models initialized with $n = 3$ can match $n = 4$ observations in the short term by tuning a viscosity multiplier, but this compensation breaks down on projection timescales [36].

2. *Missing calving.* No ISMIP6 model adequately represents iceberg calving [32, 56], which is a first-order control on grounding-line retreat rate. The sensitivity of glacier flow to major calving events is well documented (e.g., 2–8× speedup after Larsen B collapse; [57]).

3. *Initialization uncertainty.* Goldberg et al. [58] showed that the choice of calibration data (velocity vs. surface elevation) produces a factor-of-two spread in projected Thwaites loss rates—comparable to the inter-model spread in ISMIP6. Surface-elevation calibration, which better captures inland thinning signatures of MISI, consistently produces larger mass loss than velocity calibration.

4. *Incomplete uncertainty quantification.* ISMIP6 integrates over model structure but not over parametric uncertainty within individual models, which alone could produce an interquartile range of up to 12.9 cm SLE for RCP8.5 at 2100 [59].

5. *Biased ensemble.* ISMIP6 models share numerical methods, parameterizations, and jointly omit key physics—they are not independent samples from the space of possible models [32].

6. *Historical validation failure.* Few if any ISMIP6 models reproduce the observed trajectory of WAIS mass loss; the ensemble 95th percentile merely tracks the lower bound of IMBIE observations [32]. Under continued warming, IPCC models show slower sea-level rise rates than extrapolation of modern observations [1].

Independent cross-check: structured expert judgment

The Bamber et al. [49] structured expert judgment (SEJ) provides an independent, methodology-orthogonal validation target. Their WAIS-specific results under +5 °C warming give a median of 180 mm and 95th percentile of 930 mm, with strong positive skew. Our A4 mixture (median ∼400 mm, 90% CI: 60–1500 mm, after rheology correction) is wider and shifted higher than the Bamber estimates. This is expected because: (a) Bamber was published before the rheology bias was quantified [36]; (b) the SEJ was conducted before Morlighem et al. [51] and recent Thwaites observations that have added to the evidence for ongoing instability; and (c) our framework explicitly applies the rheology correction, which shifts the entire distribution upward by ∼28%. The agreement in the broad distributional shape—strongly right-skewed, with median in the hundreds of millimeters—strengthens confidence in both approaches despite their different methodologies.

As a second cross-check, Rosier et al. [60] calibrated an ice sheet model (Úa) against ASE observations using Bayesian inference with deep surrogates and projected Amundsen Sea sector contributions of 19.0±2.2 mm (Paris 2 °C) and 18.9±2.7 mm (RCP8.5) by 2100—statistically indistinguishable across scenarios, supporting the SSP-independence assumption adopted here. Their calibrated contribution falls within our S1 range (25–85 mm), consistent with the expectation that process models calibrated against the short observational record produce estimates at the lower end of our framework. Rosier et al. include both cliff-height- and hydrofracture-driven calving but find no evidence of unstable calving-front retreat in any simulation, and their creep-exponent prior ($n \in [2, 5]$, uniform) is broad but centered lower than the satellite-derived $n = 4.1 \pm 0.4$ [33]. Their narrow uncertainty range (∼5 mm, 1σ) reflects the structural constraints of a single model rather than the epistemic uncertainty that our scenario mixture model is designed to represent.

Sensitivity to scenario weights

To assess the robustness of the A4 mixture to the assigned scenario weights, we perturb each weight by ±10 percentage points (redistributing the change proportionally among the remaining scenarios) and examine the effect on the mixture median at 2100 (Fig. S11). Perturbing the S2 (MISI) weight shifts the median by less than ±0.002 m, reflecting S2's dominance at 80%. Perturbations to S1 and S3 produce larger shifts (up to ±0.05 m) but in opposite directions, so that the median remains within a ∼0.09 m range across all perturbations—still small relative to the 90% credible interval (∼1.4 m). The insensitivity to weight perturbations reflects the dominance of within-scenario uncertainty (controlled by the endpoint ranges and skewness) over the between-scenario weighting.

Repeating this analysis for the 95th percentile (5% exceedance probability)—which represents the upper tail of the distribution—yields a similar conclusion (Fig. S12). The largest sensitivity is again to the S3 weight, which shifts the 95th percentile by approximately ±12%, while S1 and S2 perturbations produce shifts of only ±3%. These variations remain a small fraction of the 95th-percentile value itself ($\sim$1.5 m), confirming that the tail behavior is governed by within-scenario uncertainty rather than the assigned weights.

1.4 MCMC convergence diagnostics

For all MCMC fits (thermosteric, glaciers, EAIS, Peninsula), we verified convergence using the Gelman–Rubin diagnostic ($\hat{R}$) and effective sample size (ESS). For the glacier and Peninsula models, all parameters satisfy $\hat{R} < 1.05$ with bulk ESS ≥ 120 and tail ESS ≥ 159; mean acceptance fractions are 0.44 (glaciers, linear) and 0.50 (Peninsula). For the EAIS diagnostic fit, all parameters satisfy $\hat{R} < 1.05$ with bulk ESS > 120 and acceptance fraction 0.49. For the thermosteric two-layer model, the mean acceptance fraction is 0.455 across 48 walkers with 576,000 posterior samples (after burn-in); convergence diagnostics are not computed via the standard $\hat{R}$ formula (the thermosteric model uses a custom MCMC loop rather than the shared level-space fitting routine), but the large sample size and stable trace plots confirm convergence. Components share a common temperature driver, introducing positive correlations in the projection ensemble. This correlation is captured sample-by-sample (each Monte Carlo realization uses the same SSP temperature trajectory for all components) but the structural uncertainty in the temperature trajectories introduces additional correlation not explicitly modeled.

1.5 Rate-space blending

The total GMSL forecast is constructed by blending the satellite-era quadratic extrapolation with the component-model sum in rate space:

$$\dot{H}_{\text{forecast}}(t) = w(t)\,\dot{H}_{\text{quad}}(t) + \big[1 - w(t)\big]\,\dot{H}_{\text{comp}}(t), \tag{16}$$

where $\dot{H}_{\text{quad}}$ is the rate from the satellite-era quadratic fit (Section "GMSL rate estimates"), $\dot{H}_{\text{comp}}$ is the rate of the component sum (finite differences of the annual-resolution component projections), and $w(t)$ is a sigmoid blending weight

$$w(t) = 1 - \sigma\!\left(\frac{t - t_c}{\tau}\right), \tag{17}$$

with $\sigma(x) = (1 + e^{-x})^{-1}$ the standard logistic function, $t_c = 2035$, and $\tau = 5$ yr. At $t \ll t_c$, $w \approx 1$ and the forecast follows the observed quadratic trajectory; at $t \gg t_c$, $w \approx 0$ and the forecast follows the component-model physics. The transition spans approximately 22 yr (10%–90% of the weight shift occurs over $\pm 2.2\,\tau \approx \pm 11$ yr around t_c).

The level forecast is obtained by integrating Eq. (16) forward from the observed GMSL at the forecast origin $t_0 = 2025.3$ (the end of the NASA altimetry record) using the trapezoidal rule:

$$H_{\text{forecast}}(t) = H_{\text{obs}}(t_0) + \int_{t_0}^{t} \dot{H}_{\text{forecast}}(t')\,dt'. \tag{18}$$

The blending is performed sample-by-sample across the 2,000-member Monte Carlo ensemble, preserving correlations between the quadratic-fit parameter uncertainty and the component-model uncertainty.

Because the quadratic extrapolation is derived entirely from the satellite altimetry record, it carries no dependence on emissions pathway. The component sum is SSP-dependent through the temperature-forced component models. Consequently, the blended forecast is SSP-independent in the near term (where $w \approx 1$) and becomes progressively SSP-dependent as the component physics dominate at longer lead times.

1.6 Budget closure diagnostic

The residual between the component sum and NASA altimetry is distributed across components in proportion to their variances: $\text{shift}_i = (\sigma_i^2 / \sum_j \sigma_j^2) \times r$, where r is the total residual and σ_i is the standard deviation of component i from its Monte Carlo ensemble (the Gaussian assumption is approximate for WAIS, whose distribution is skewed; however, the diagnostic is applied to the hindcast period where the WAIS contribution is small and approximately linear). This is the Bayesian posterior update under joint Gaussian assumptions, applied as a diagnostic without modifying the component posteriors. The shift-to-sigma ratio for each component tests whether the implied adjustment is within the stated uncertainty.

1.7 Variance decomposition

A variance decomposition using the law of total variance across the three policy-relevant SSPs (1-2.6, 2-4.5, 3-7.0) attributes 92% of total forecast variance to within-scenario uncertainty—predominantly WAIS (91% of total variance)—and 8% to between-scenario differences. Cross-component covariances are small (+0.0014 m^2, representing 0.5% of total variance), confirming that the independent calibration approach does not introduce material correlations.

1.8 Temperature scenarios

SSP temperature trajectories (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5) are from IPCC AR6 assessed warming projections. Historical forcing uses Berkeley Earth monthly GMST [4]. All anomalies are relative to a 1995–2005 baseline (≈2000 reference year). Forecasts extend from 2020 to 2150, with pre-2020 values from the observational record. SSP GMST trajectories are converted to Greenland regional surface temperature using an Arctic amplification factor of 3.0 ± 0.5 (1σ), consistent with the observed ratio of Greenland-to-global warming over 1970–2021.

1.9 Prior specifications

Table S2 lists the prior distributions used in each Bayesian component model. All priors are weakly informative and are chosen to regularize the posterior without dominating the likelihood. All values in the table are reported in mm (the code operates in meters internally).

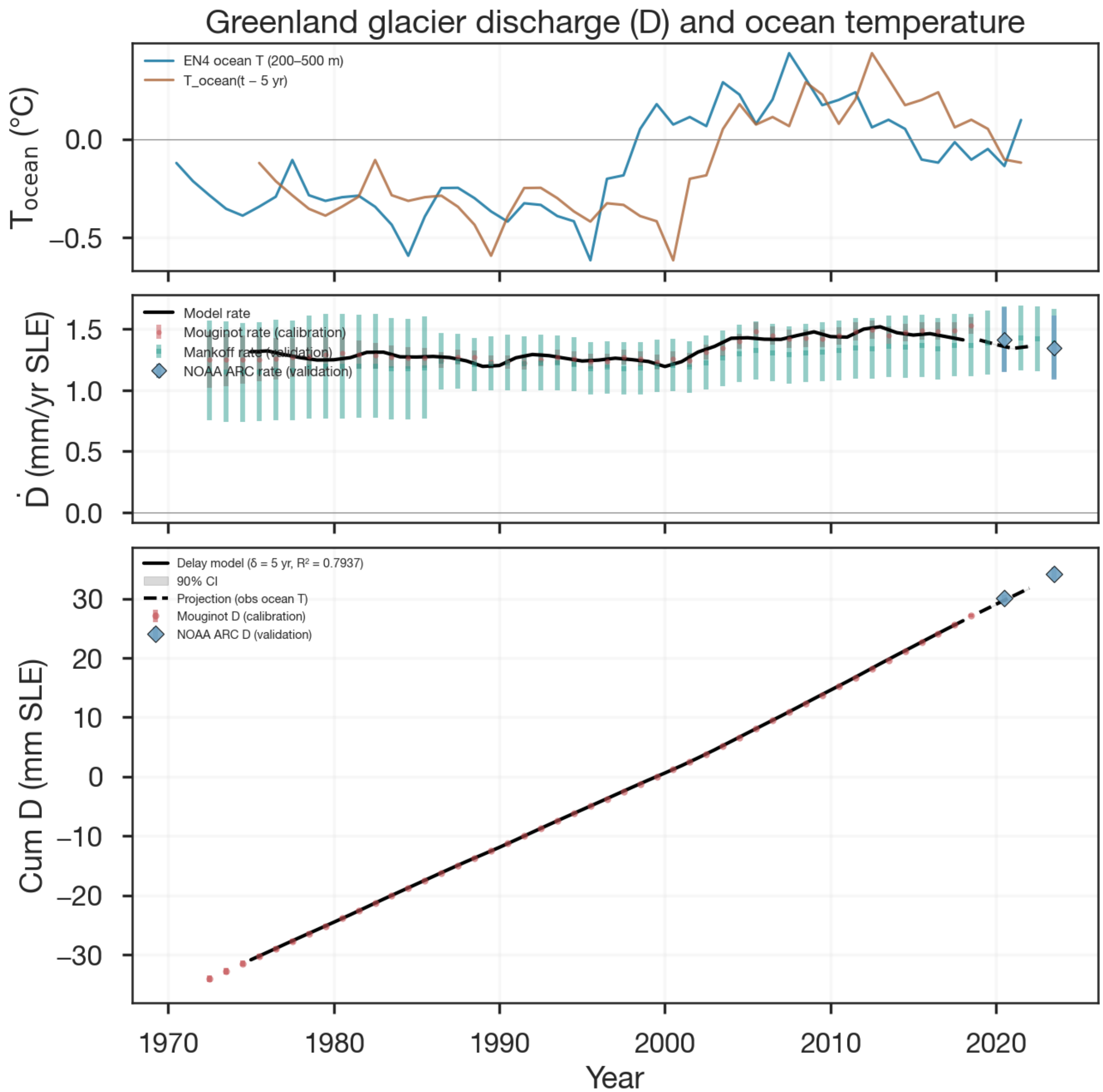

Figure S4: **Greenland discharge delay model fit.** (a) EN4 subsurface ocean temperature (200–500 m, annual) and the delayed signal $T_{\text{ocean}}(t-\delta)$ with $\delta = 5\,\text{yr}$. (b) Discharge rate anomaly: Mouginot (calibration), Mankoff (validation), and NOAA Arctic Report Card (validation) observations compared with the model rate and 90% posterior CI. (c) Cumulative discharge: model fit (solid, calibration period) and forward projection (dashed), with Mouginot calibration data and NOAA ARC validation data.

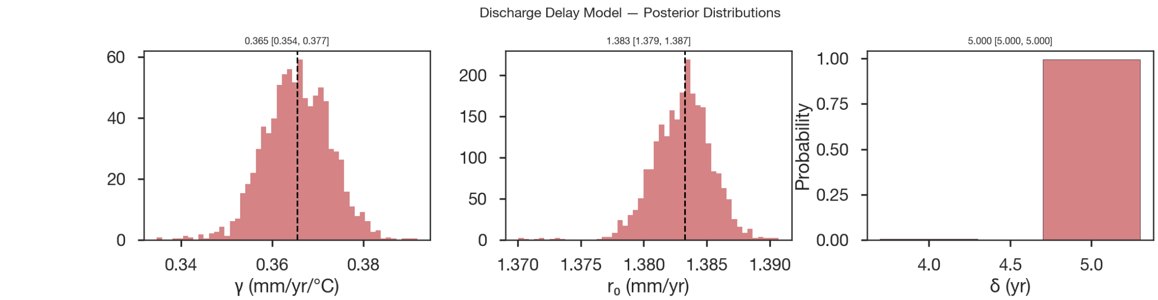

Figure S5: **Greenland discharge model posteriors.** Posterior distributions for the three parameters of the delay model: ocean thermal forcing sensitivity γ, background rate r_0, and time delay δ. The delay δ is discrete (BIC-weighted mixture over $\delta \in \{4, 5, 6, 7, 8\}$ yr); the remaining parameters are continuous.

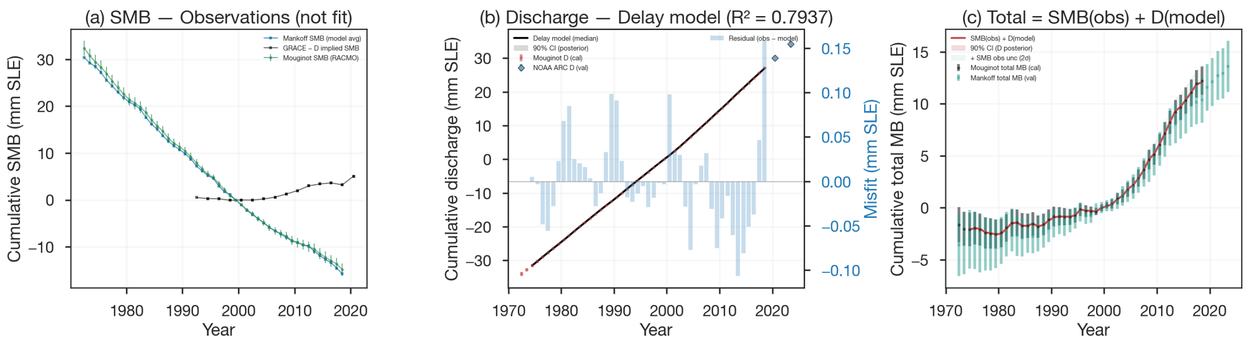

Figure S6: **Greenland SMB, discharge, and total mass balance diagnostics.** (a) Cumulative SMB from Mouginot (RACMO-derived), Mankoff, and GRACE − D implied SMB. SMB is used as observations, not fit to temperature. (b) Cumulative discharge from the delay model with 90% posterior CI, validated against Mouginot (calibration) and NOAA ARC (validation). (c) Total Greenland mass balance: SMB(obs) + D(model) compared with Mouginot (calibration) and Mankoff (validation) total mass balance records.

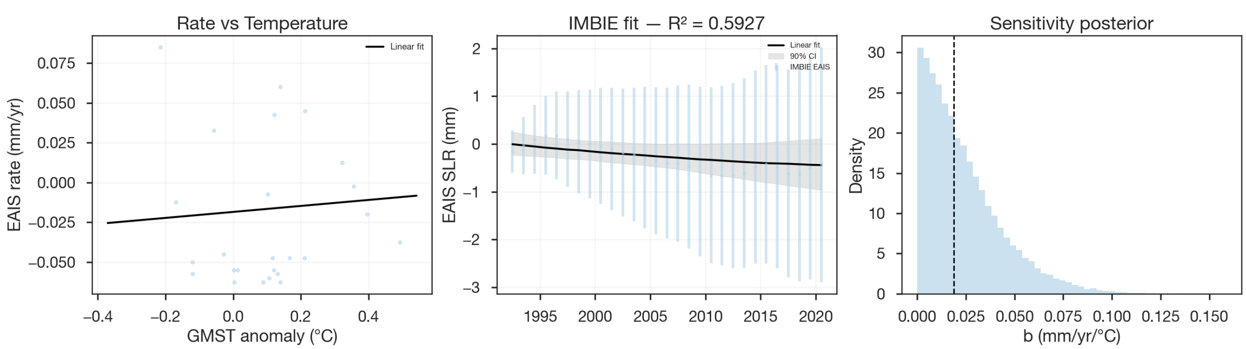

Figure S7: **EAIS model diagnostics.** (a) EAIS mass loss rate vs GMST anomaly with the fitted quadratic relationship. (b) Cumulative EAIS SLR: IMBIE observations with the model fit and 90% credible interval. (c) Posterior distribution for the temperature sensitivity b.

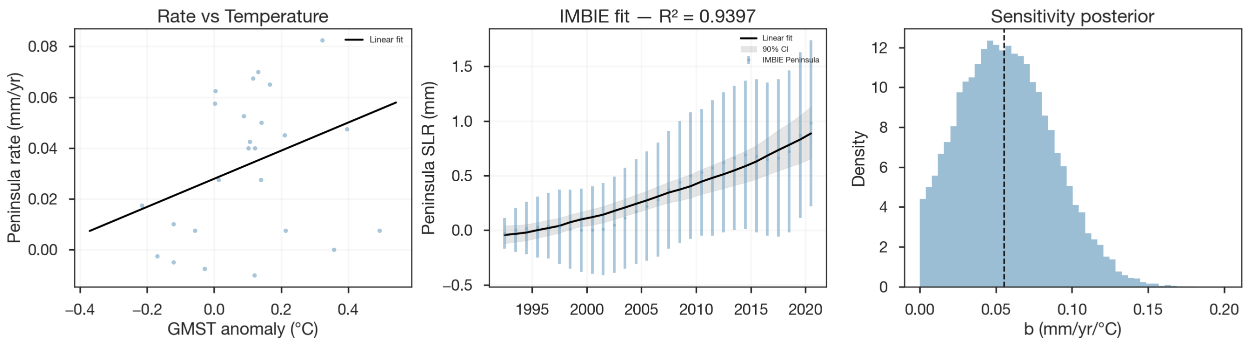

Figure S8: **Peninsula model diagnostics.** (a) Peninsula mass loss rate vs GMST anomaly with the fitted linear relationship. (b) Cumulative Peninsula SLR: IMBIE observations with the model fit and 90% credible interval; $R^2 = 0.94$. (c) Posterior distribution for the temperature sensitivity b.

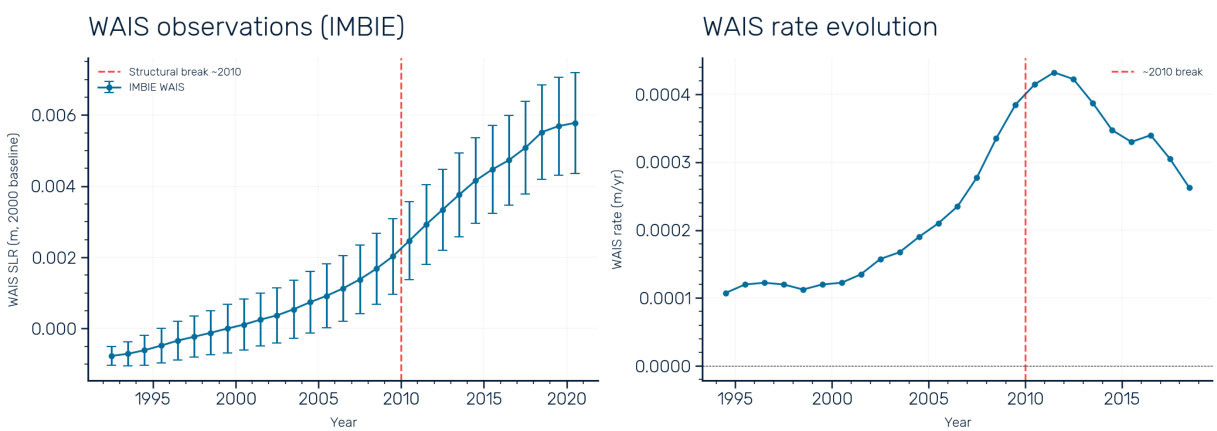

Figure S9: **WAIS observational record.** Left: cumulative WAIS sea-level contribution from IMBIE (1992–2020; [16]) with measurement uncertainties, relative to the year-2000 baseline. The structural break at ∼2010 (red dashed line) separates a period of acceleration from apparent deceleration. Right: WAIS mass loss rate evolution, showing the rate increasing from ∼0.15 mm yr^{-1} in the early 1990s to a peak of ∼0.44 mm yr^{-1} around 2010–2012, followed by a decline that may reflect internal ocean variability in the Amundsen Sea [38].

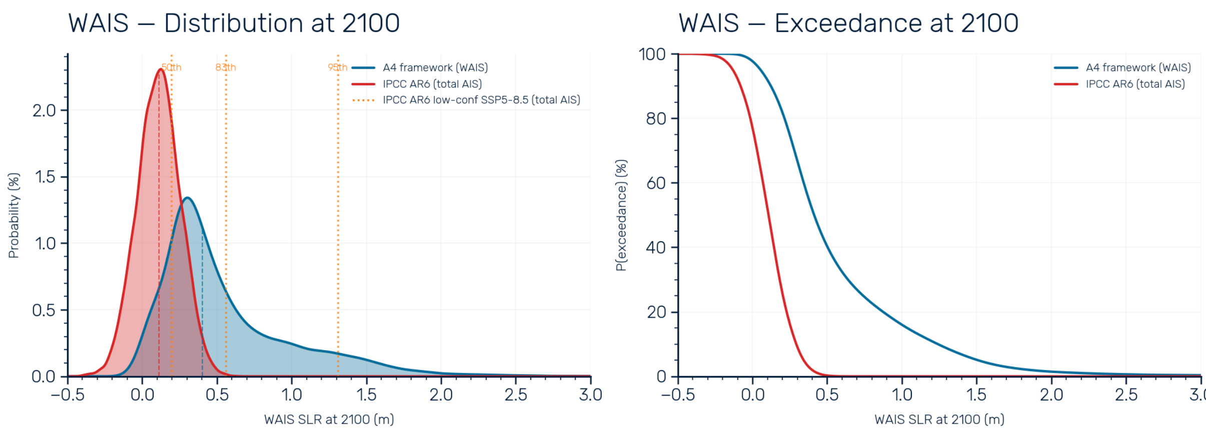

Figure S10: **WAIS endpoint distributions and exceedance probabilities: A4 vs IPCC.** Left: probability density at 2100 for the A4 WAIS framework (blue) compared with the IPCC AR6 total Antarctic projection (red). The A4 distribution is wider, with a higher median and a heavier right tail arising from the MISI skewness and rheology correction. Right: exceedance probability curves, showing that the A4 framework assigns substantially higher probability to exceeding any given threshold above ∼0.1 m. The IPCC distribution assigns near-zero probability to outcomes exceeding 0.5 m, whereas the A4 framework assigns ∼40% probability to this threshold.

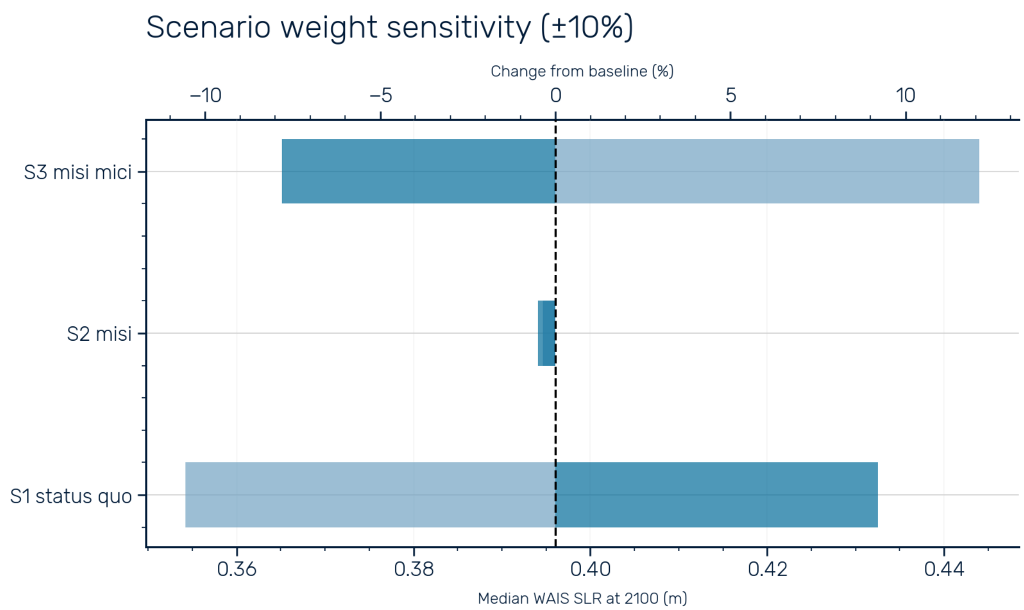

Figure S11: **Scenario weight sensitivity.** Tornado plot showing the effect of ± 10 percentage-point perturbations to each scenario weight on the median WAIS SLR at 2100. Dark bars show the inner perturbation range; light bars extend to the full $\pm 10\%$ range. The dashed vertical line indicates the baseline median (0.40 m). The mixture median is insensitive to weight perturbations, varying by less than 0.09 m across all perturbations—small relative to the 90% credible interval of the mixture distribution ($\sim$1.4 m).

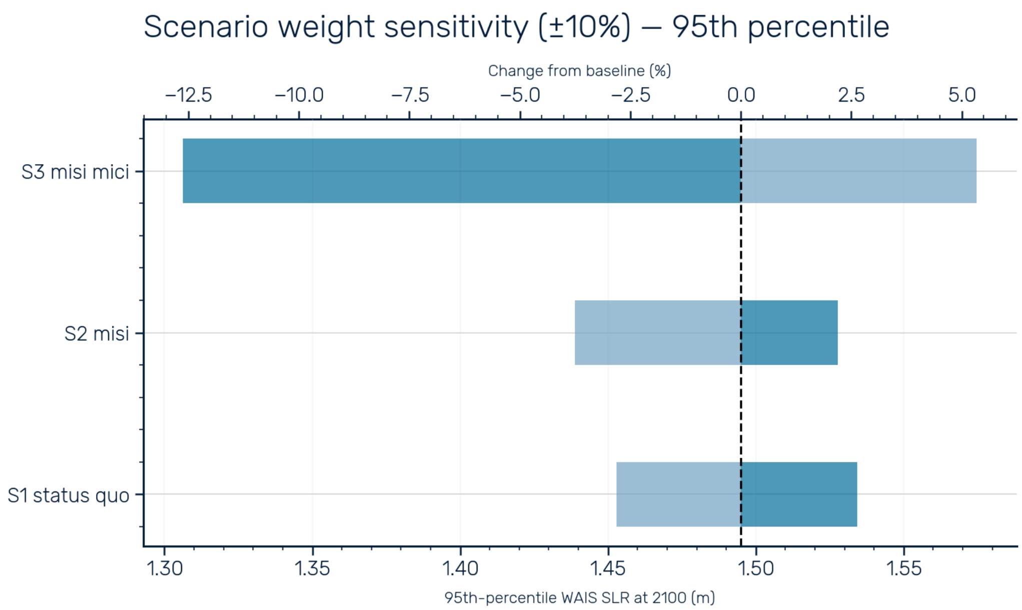

Figure S12: **Scenario weight sensitivity—95th percentile.** As in Fig. S11, but for the 95th percentile (5% exceedance) of the WAIS SLR distribution at 2100. The upper axis shows the percentage change from the baseline value (∼1.5 m). The tail of the distribution is similarly insensitive to weight perturbations, with the largest shift ($\sim \pm$ 12% from S3) remaining a small fraction of the total.

Table S2: Prior distributions for all component models. Exp = Exponential, HN = Half-Normal, HC = Half-Cauchy, LN = Log-Normal, N = Normal, $\mathrm{SN}_{\log}$ = Skew-Normal in log-space. Parameters a, b, c in the level-space models correspond to the quadratic, linear, and constant coefficients of the rate–temperature relationship, respectively.

Component	Parameter	Prior	Source / justification
Thermosteric (two-layer ODE, 6 free parameters)			
	a (quadratic)	$\mathrm{Exp}(\mu = 22$ mm/°C$^2)$	Positive curvature; regularizes toward linear
	b_u (upper sensitivity)	$\mathrm{HN}(\sigma = 150$ mm/°C$)$	Weakly informative; encompasses literature range
	b_d (deep sensitivity)	$\mathrm{HN}(\sigma = 100$ mm/°C$)$	Weakly informative; smaller than upper layer
	c (secular trend)	$\mathrm{N}(0.3, 0.5^2)$ mm yr^{-1}	Centered near zero; allows small drift
	$\log \tau_u$	$\mathrm{N}(\log 8, 0.5^2)$	LN median 8 yr; Geoffroy et al. range
	σ_{extra}	$\mathrm{HC}(\gamma = 3$ mm$)$	Model inadequacy; weakly informative
Glaciers, EAIS, Peninsula (level-space Bayesian, 5 free parameters each)			
	a $(d\alpha/dT)$	$\mathrm{Exp}(\mu = 0.043$ mm yr^{-1} °C$^{-2})$	PC prior; $P(a > 0.10) = 10\%$
	b (α_0)	$\mathrm{HN}(\sigma = 0.087$ mm yr^{-1} °C$^{-1})$	Weakly informative; encompasses literature range
	c (trend)	$\mathrm{N}(0.3, 1.0^2)$ mm yr^{-1}	Weakly informative; centered on observed rate
	σ_{extra}	$\mathrm{HC}(\gamma = 5$ mm$)$	Model inadequacy
	H_0 (baseline offset)	$\mathrm{N}(H_{\mathrm{obs}}(t_1), 5^2)$ mm	Centered on first observation
Greenland SMB (literature sensitivities, not MCMC)			
	C_T (linear)	$\mathrm{N}(-300, 80^2)$ Gt yr^{-1} °C^{-1}	Inter-RCM spread (MAR, RACMO, HIRHAM)
	C_{T^2} (quadratic)	$\mathrm{N}(-50, 30^2)$ Gt yr^{-1} °C^{-2}	RCM nonlinearity estimates
Greenland discharge (WLS, no explicit priors)			
	γ, r_0	WLS point estimates	Implicit uniform prior via least squares
	δ (delay)	BIC-weighted mixture	$\delta \in \{4, 5, 6, 7, 8\}$ yr; data-driven
WAIS (scenario mixture, 3 scenarios)			
	$P(\mathrm{S1})$	0.10	Assessed: continued present-day rates
	$P(\mathrm{S2})$	0.80	Assessed: MISI-driven acceleration
	$P(\mathrm{S3})$	0.10	Assessed: MISI + cliff instability

Continued on next page

Table S2 continued from previous page

Component	Parameter	Prior	Source / justification
	S1 endpoint	$\mathrm{SN}_{\log}$, 25–85 mm	IMBIE extrapolation
	S2 endpoint	$\mathrm{SN}_{\log}(\alpha = 4)$, 150–1000 mm	Goldberg et al. lower bound; right-skewed
	S3 endpoint	$\mathrm{SN}_{\log}(\alpha = -3)$, 600–1300 mm	MISI + MICI; AR6 low-confidence AIS p83–p95
	S2 trajectory β	$\mathrm{LN}(\ln 1.8, 0.3^2)$	MISI acceleration; Schoof (2007) scaling
	S3 trajectory β	$\mathrm{LN}(\ln 2.2, 0.3^2)$	MISI + MICI; more back-loaded
	n_{draw} (Mode B)	$\mathrm{N}(4.1, 0.4^2)$, $n \geq 3$	Millstein et al.; Ranganathan & Minchew
	r_0 (rheology sensitivity)	0.28	Martin et al.; $R(n) = 1 + r_0\,(n - 3)$

1.10 Budget closure and validation

Each component is calibrated independently; the total sea level is a diagnostic sum, not a calibration target. We evaluate budget closure by comparing the temperature-forced component hindcast (excluding EAIS; see main text) against NASA satellite altimetry over three epochs (Table S3).

Table S3: Budget closure: component-sum rate versus NASA GMSL rate over three epochs. All rates from linear fits to the temperature-forced hindcast (no EAIS) and NASA altimetry.

Epoch	NASA (mm yr^{-1})	Sum (mm yr^{-1})	Residual (mm yr^{-1})	Closure
1993–2025	3.29	3.21	0.09	97.4%
2000–2025	3.49	3.53	−0.04	101.0%
2010–2025	4.11	3.83	0.28	93.3%

A variance-weighted attribution distributes the residual across components in proportion to their uncertainties (Fig. S13). Terrestrial water storage absorbs the largest share (45%), reflecting poorly characterized structural uncertainty in groundwater and reservoir reconstructions. The thermosteric component absorbs 31%, consistent with the residual structural uncertainty (5%) retained after the Li et al. [8] bias correction. All components are within 0.4 standard deviations of their independent estimates. The framework closes the observed budget to within 3% over 1993–2025 and 7% over 2010–2025 without requiring any component to depart from what the observations support.

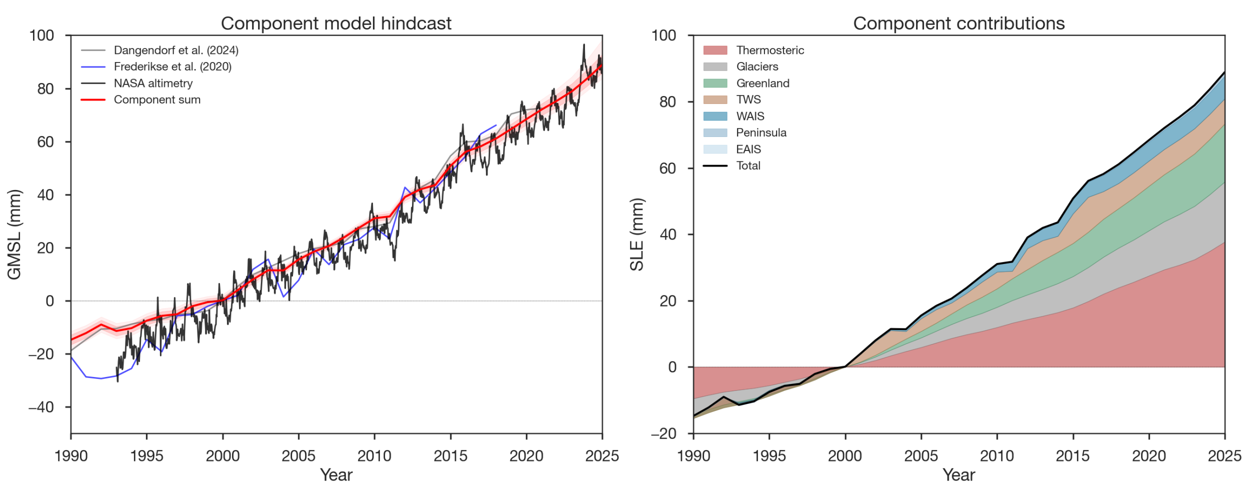

Figure S13: **Budget closure: temperature-forced component hindcast versus observed GMSL. (A)** Total component sum (red, with EAIS; orange dashed, without EAIS) compared with NASA altimetry (black), Frederikse et al. (blue), and Dangendorf et al. (grey) GMSL reconstructions. **(B)** Per-component contributions to the hindcast. All values in mm relative to the year 2000 baseline.

2 Value of Information Analysis

The value of information (VoI) analysis quantifies the economic benefit of resolving the binary uncertainty about whether WAIS retreat remains slow (Scenario 1) or allows for the possibility of rapid retreat (weighted mixture of Scenarios 1–3). The framework follows the expected value of sample information (EVSI) formulation from classical decision theory [61].

2.1 Decision framework

We consider an illustrative coastal adaptation decision where a planner must choose a defense height d (meters of sea-level protection) to minimize total expected costs. For a given sea-level rise distribution S at 2100, the total annual cost of defense height d is

$$C(d) = C_{\text{adapt}}(d) + \mathbb{E}\big[D\big(\max(S - d,\, 0)\big)\big], \tag{19}$$

where $C_{\text{adapt}}(d) = c_a \cdot d$ is the annualized adaptation cost and $\mathbb{E}\big[D(\cdot)\big]$ is the expected value of the residual damage D from SLR exceeding the defense height (overtopping).

2.1.1 Adaptation cost

We estimate the global adaptation cost per meter of GMSLR as $c_a = \$4\,\mathrm{T/yr/m}$. This is derived from Tiggeloven et al. [62], who estimate that 3.4% of the global coastline ($\sim$54,400 km of $\sim$1.6 million km total) warrants dike heightening under cost-benefit optimality, at a unit cost of \$7 M per km per meter of dike height. This yields $\sim$\$400 B/m for dike raising alone along the optimal 3.4% of coastline. We apply a conservative $10\times$ multiplier to account for the full societal adaptation burden—including seawalls, nature-based solutions, managed retreat, and infrastructure relocation—across a broader fraction of the global coastline.

2.1.2 Damage function

Residual flood damages are computed using the piecewise-linear global damage function of Jevrejeva et al. [63], which maps SLR to annual global flood costs based on estimated asset exposure. For SLR beyond the tabulated range (1.80 m), we extrapolate linearly using the slope of the final segment ($\sim$\$13.8 T/yr per meter).

2.2 Two-world structure

The forecast distribution is decomposed into two conditional distributions:

- **Slow WAIS** (S_{slow}) samples from the blended forecast conditional on WAIS Scenario 1 representing future GMSLR.
- **Fast WAIS** (S_{fast}) samples from the full (unconditional) forecast that exceed the 97.5th percentile of the stable distribution, representing outcomes that allow for rapid WAIS retreat (Scenarios 2–3).

The current prior probability of stable WAIS is $p_s = 0.10$, consistent with our scenario weights (Scenario 1 receives 10% probability).

2.3 EVSI calculation

For a given prior $p_s = P(\text{slow WAIS})$, the EVSI quantifies the annual benefit of learning whether WAIS will lose mass slowly or rapidly before choosing the defense height for coastal development.

2.3.1 Hedging cost (no information)

Without additional information, the planner hedges by choosing the single defense height d^*_{hedge} that minimizes the prior-weighted expected cost:

$$C^*_{\text{hedge}} = \min_d \left[p_s \cdot C_{\text{slow}}(d) + (1 - p_s) \cdot C_{\text{fast}}(d) \right], \tag{20}$$

where $C_{\text{slow}}(d) = C_{\text{adapt}}(d) + \mathbb{E}[D(\max(S_{\text{slow}} - d, 0))]$ and $C_{\text{fast}}(d)$ is defined analogously using S_{fast}.

2.3.2 Perfect information cost

With perfect information, the planner would tailor the defense height to whichever world is realized:

$$C^*_{\text{perfect}} = p_s \cdot \min_d C_{\text{slow}}(d) \; + \; (1 - p_s) \cdot \min_d C_{\text{fast}}(d). \tag{21}$$

2.3.3 Annual EVSI

The annual value of perfect information is the difference between the hedging cost and the perfect-information cost:

$$\text{EVSI}_{\text{ann}} = \max\bigl(C^*_{\text{hedge}} - C^*_{\text{perfect}}, \; 0\bigr). \tag{22}$$

This is the annual cost penalty of having to compromise on a single design rather than tailoring to the true state. When $p_s \approx 0$ or $p_s \approx 1$, the hedging design is already close to the tailored design, so EVSI is small. EVSI is maximized near $p_s \approx 0.5$, where the optimal designs for the two worlds diverge most.

2.4 Discounting and the VoI surface

The annual EVSI is converted to a net present value (NPV) by discounting the stream of benefits from the year information arrives (t_a) to the target year ($t_T = 2100$) at a real discount rate $r = 3\%$:

$$\text{VoI}(t_a, p_s) = \text{EVSI}_{\text{ann}}(p_s) \left(\frac{1 - (1 + r)^{-(t_T - t_a)}}{r} \right). \tag{23}$$

The annuity factor $[1 - (1 + r)^{-n}]/r$ captures the number of discounted years over which the improved decision yields benefits.

The VoI surface (Fig. 4c of the main text) is computed over a grid of arrival years ($t_a \in [2026, 2094]$) and prior probabilities ($p_s \in [0.01, 0.99]$). All calculations use the 3 °C

(SSP2-4.5) warming trajectory. The choice of warming trajectory has qualitatively little impact on the results given all the uncertainties involved. Note that our cost estimates are likely to be underestimates, but the takeaway message remains even if the true costs are an order of magnitude lower.

2.5 Key assumptions and limitations

The VoI calculation is intended to be simple analysis to make a simple point, not to stand for detailed financial decisions. As such, we make several simplifying assumptions:

1. **Binary resolution.** Information perfectly resolves the slow/fast dichotomy. Partial information (e.g., updating p_s from 0.10 to 0.30 without full resolution) would yield lower VoI.

2. **Single decision.** The planner makes one irreversible design choice. In practice, adaptive management with staged decisions would reduce the hedging penalty, lowering VoI.

3. **Global aggregation.** Costs are aggregated globally. Local VoI would differ based on coastal exposure and elevation.

4. **Linear adaptation cost.** We assume C_{adapt} scales linearly with defense height. Economies or diseconomies of scale would modify the optimal designs.

5. **Discount rate.** The 3% real discount rate is standard for infrastructure but lower rates (reflecting intergenerational equity) would increase VoI. Higher discount rates, that would lower VoI, are unlikely given the long timeline of risk exposure from sea level rise.

Despite these simplifications, the analysis provides a rough order-of-magnitude estimate of the economic value of resolving WAIS uncertainty, supporting the argument that targeted investment in ice-sheet observations and process understanding has massive expected returns.

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